All-orders torus structure and affine-linear local phase recurrences in Type-1, $N = 3$ Landau–Zener

OpenAI working draft
(Dated: April 18, 2026)

At a simple transport ramification point of a Type-1, $N = 3$ Landau–Zener family, we show that the canonical exact two-sheet gauge is torus-valued to all orders. Ordinary Lagrange–Bürmann inversion then yields a compact closed recurrence for the odd local phase coefficients $D_{2m+1}(H)$, and because the commuting-family member enters only through the linear factor L_H, the full member-dependent phase series is affine-linear in $L_H(v)$ and $L'_H(v)$ at every order. In the WKB-normalized Airy convention the local selector block therefore reduces to a scalar renormalization of the universal Airy jump. To descend this exact two-sheet theorem back to the original outer matcher, we state two narrow hypotheses supported by high-precision large-sample assays. The remaining open content is then concentrated in the local activation–multiplier morphism: selector sufficiency and the exact global insertion rule for the scalar local block.

Setup and goal. The project begins with the gauged adiabatic Type-1 evolution

$$f'_i(u) = -\sum_{j\neq i} S_{ij}(u) f_j(u) + \mathrm{i}\big(E_i(u) - E_x(u)\big) f_i(u), \quad (1)$$

and its quotient-ring reduction

$$\partial_u[R] = \mathrm{i}\big[(m - E_x p) s_u R\big], \qquad \mathcal{E}_u = \mathbb{C}[\lambda]/(q_u). \quad (2)$$

Here $q_u(\lambda) = up(\lambda) - n(\lambda)$. with commuting triple (H_x, Y, R) and common/spinor cover geometry already formalized in Refs. 1–6. The unresolved local content was concentrated in the matcher $G_{v,s}(H)$ at a simple transport ramification point. Here we show that the exact canonical two-sheet reduction already forces a torus-valued local gauge, derive an all-orders phase recurrence, and isolate precisely what remains open.

Exact two-sheet torus theorem. Let v be a simple transport ramification point, $u_* = u(v)$, and write $u = u_* + t^2$. If $\lambda_\pm(t)$ are the two colliding local branches, then the reduced sheet carriers satisfy exact scalar equations

$$\partial_u R_\pm = \mathrm{i}(E_\pm - E_x) R_\pm, \quad (3)$$

so on the t-cover the active two-sheet system is exactly diagonal,

$$\partial_t \begin{pmatrix} R_+ \\ R_- \end{pmatrix} = \big(A_v(t;H) I + B_v(t;H)\sigma_3\big) \begin{pmatrix} R_+ \\ R_- \end{pmatrix}. \quad (4)$$

After reconstructing the half-form amplitudes,

$$\Psi_\pm(t;H) = \sqrt{2t}\,\Gamma_\pm(t) R_\pm(t;H)\sqrt{\mathrm{d}t}, \quad (5)$$

and removing the common scalar phase, the exact basis is diagonal in the ordered resolved-sheet splitting $L_+ \oplus L_-$. The WKB-normalized Airy basis preserves the same ordered splitting. The change-of-basis matrix between the two therefore preserves each line separately, hence is diagonal. After determinant-one normalization,

$$\widehat{G}_v(t;H) = \exp\!\big(\Theta_v(t;H)\sigma_3\big). \quad (6)$$

This is an exact theorem for the canonical two-sheet reduction: no torus-breaking off-diagonal term can survive in that normalization.

Lagrange–Bürmann kernel. Write $z = \lambda - v$ and factor the local map as

$$u(v+z) - u_* = z^2 h_v(z), \qquad h_v(0) = \frac{u''(v)}{2} \neq 0. \quad (7)$$

Define $\phi_v(z) = z\sqrt{h_v(z)}$ and let ψ_v be its ordinary local inverse. Then

$$\lambda_+(t) = v + \psi_v(t), \qquad \lambda_-(t) = v + \psi_v(-t). \quad (8)$$

With $F_H(z) = \Delta_H(v+z)$ and $\Delta_H(\lambda) = \mu(\lambda)L_H(\lambda)/p(\lambda)$, define

$$\Delta_H(\lambda_+(t)) - \Delta_H(\lambda_-(t)) = 2\sum_{m\geq 0} D_{2m+1}(H) t^{2m+1}. \quad (9)$$

Ordinary Lagrange–Bürmann inversion gives the all-orders kernel

$$D_{2m+1}(H) = \frac{1}{2m+1}[z^{2m}]F'_H(z)\left(\frac{z}{\phi_v(z)}\right)^{2m+1} = \frac{1}{2m+1}[z^{2m}]F'_H(\quad (10)$$

Writing $h_v(z) = h_0(1 + \sum_{k\geq 1} a_k z^k)$ and $Q_m(z) = (1 + \sum_{k\geq 1} a_k z^k)^{-(2m+1)/2} = \sum_{n\geq 0} q_n^{(m)} z^n$, one obtains the recursive coefficients

$$q_n^{(m)} = \frac{1}{n}\sum_{k=1}^{n}\left(\frac{1-2m}{2}k - n\right) a_k q_{n-k}^{(m)}. \quad (11)$$

Affine-linearity on the commuting family. For Type-1, $\Delta_H = FL_H$ with $F(\lambda) = \mu(\lambda)/p(\lambda)$ and L_H linear. Hence $L_H(v+z) = L_H(v) + L'_H(v)z$, so every odd phase coefficient splits as

$$D_{2m+1}(H) = A_m(v)L_H(v) + B_m(v)L'_H(v), \quad (12)$$

with geometry-only sequences $A_m(v)$ and $B_m(v)$. The full member-dependent phase series is therefore an explicitly computable affine-linear sequence on the whole commuting family. In particular, if

$$\Theta_v^{\mathrm{ph}}(t;H) = \sum_{n\geq 2} \Xi_{2n+1}(H) t^{2n+1}, \quad (13)$$

then

$$\Xi_{2n+1}(H) = \frac{2\mathrm{i}}{2n+1} D_{2n-1}(H), \qquad n \geq 2. \tag{14}$$

The first new explicit coefficient beyond the previously derived t^5 term is thus $\Xi_7(H) = \frac{2\mathrm{i}}{7} D_5(H)$; explicit formulas for Ξ_7 and Ξ_9 for the canonical v_4 point are given in the Supplement.

Local scalar multiplier. Because $\mathrm{d}u = 2t\,\mathrm{d}t$, the exact local phase mismatch is

$$S_{+-}(t;H) = 4 \sum_{m\geq 0} \frac{D_{2m+1}(H)}{2m+3} t^{2m+3}. \tag{15}$$

The linear Langer coordinate removes the cubic term exactly, so the local phase remainder is analytic. In the WKB-normalized Airy convention, the same-radius active $S_1 \to S_0$ crossing therefore carries the Borel-summed scalar multiplier

$$\mathfrak{s}_v^{\mathrm{BS}}(t;H) = -\mathrm{i}\exp\bigl(2\Theta_v(t;H)\bigr). \tag{16}$$

Thus the local selector data in the canonical two-sheet normalization are scalar rather than matrix-valued.

Numerically supported descent back to the outer frame. To identify the exact two-sheet theorem with the original outer matcher $G_{v,s}(H)$, we use two narrow hypotheses: (H1) after spectator decoupling and curvature subtraction, no nondecaying torus-breaking off-diagonal survives; (H2) the residual constant non-torus part is sector-independent and geometry-only, so it may be absorbed into the resolved-sheet convention. The empirical support is strong. A resolved-sheet probe of $\widehat{G}_v$ found no off-diagonal counterexample in 1200/1200 accepted 80-digit samples. The spectator-decoupled torus-survival assay gave 1001/1001 passes, including 301 deliberately targeted high-curvature cases; all 9 legacy high-curvature failures from the raw assay also passed on retest. The sector-independence assay gave 999/1001 passes in the base 80-digit run, with only 2 borderline extreme-curvature misses, and both passed a refined 100-digit retest. Finally, the independent recurrence validation gave 1200/1200 passes at a 10^{-2} relative threshold and 1198/1200 at 10^{-3} for (D_1, D_3, D_5) simultaneously. We therefore regard (H1)–(H2) as narrow numerically supported statements rather than theorem-level inputs.

What remains open. The present work does not yet solve the local activation–multiplier morphism. Two pieces remain: (i) a proof of selector sufficiency, i.e. that every simple phase-active transport ramification point actually inserts the scalar block (16), and (ii) the exact global insertion rule that places this scalar local block inside the continuation functor. The highest-leverage symbolic next steps are therefore to derive an all-orders recurrence for the transport exponent $\Theta_v^{\mathrm{tr}}(t)$ and to prove a local-to-global exact-WKB comparison theorem for the scalar block. Numerics should now be used only to sharpen theorem statements, especially in the extreme-curvature and nongeneric-collision regimes.

[1] *Multistate Landau–Zener Solver: Augmented ODE with Gauged Dynamics* (internal summary).

[2] *A note on the sextic gap polynomial for $N = 3$ Type-1 matrices* (internal note).

[3] *Type-1, $N = 3$ Landau–Zener primer: double cover, reduced ODE, gluing theorem, and selector* (internal primer).

[4] *Formalized strategy diagram for Type-1, $N = 3$ Landau–Zener* (internal note).

[5] *Local selector matching at a transport ramification point in Type-1, $N = 3$* (internal note).

[6] *Transport versus phase in the local gauge expansion at a simple transport ramification point* (internal note).